

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Review Algebra

1. Equations

a. Determine the solutions to the following equations.

i. $2x^2 + x - 1 = 0$

iii. $x^2 - 2x = -1$

ii. $x^2 + 3x = 0$

iv. $3x^2 + 1 = 0$

b. The following equations are false. Explain why and provide a counterexample.

i. $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

iii. $\frac{x^2 - 9}{x - 3} = x - 3$

ii. $(x + y)^2 = x^2 + y^2$

iv. $\frac{x}{x - y} = \frac{1}{y}$

2. System of Equations

- a. For each system of equations, solve for the values of x and y that satisfy both equations. Then, describe the geometry of the solution(s).

i.
$$\begin{aligned} 2x + y &= 10 \\ 3x - 2y &= 5 \end{aligned}$$

ii.
$$\begin{aligned} 2x + 4y &= 6 \\ x + 2y &= 3 \end{aligned}$$

- b. Express the rational function $f(x) = \frac{2x^3 + 3x^2 + 4x + 5}{(x - 1)^2(x^2 + 1)}$ as a sum of partial fractions.